

Raymond (Mingguang) Yang

Philadelphia, PA | Willing to relocate

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EDUCATION BACKGROUND

University of Pennsylvania, Philadelphia, PA

Aug 2024 – May 2026 (Expected)

M.S.E. in **Mechanical Engineering** (Robotics and Mechatronics stream),

GPA **3.76/4.00**

University of Toronto, Toronto, Canada

Sep 2019 – June 2024

B.A.Sc. with Honors in **Mechanical Engineering** (Mechatronics and Manufacturing stream),

GPA **3.77/4.00**

SKILLS & QUALIFICATIONS

Mechanical & CAD: SolidWorks (advanced), CATIA, GD&T, mechanism design, tolerance stack-up, DFM

Mechatronics Systems: motors (servo/stepper), sensors, power systems, wiring harnesses, PCB integration

Embedded & Controls: C/C++, Arduino/ESP32/Rpi, PWM motor control, encoder feedback, CAN/I2C/SPI

Robotics Stack: ROS/ROS2, Python, Linux, hardware bring-up, system integration, debugging

Prototyping & Build: CNC machining, 3D printing, rapid prototyping, assembly, hands-on testing and iteration

ROBOTICS EXPERIENCES

Multi-Platform Lunar Rover System (NASA LuSTR), UPenn, Philadelphia, PA

Jan 2025 – Apr 2026

- Designed an aluminum rover chassis and docking structure in SolidWorks, supporting 60 kg payload while optimizing manufacturability, fit, and structural reliability for field deployment.
- Built and iterated physical prototypes using CNC-machined and 3D-printed parts, resolving mechanical integration issues across multiple design-build-test cycles.
- Designed interface PCBAs and wiring harnesses for power, sensing, and actuation, and led hardware bring-up of motors, encoders, IMU, and 48V/12V power distribution.
- Developed the actuator and sensor control stack in C++/ROS2 on Raspberry Pi, implementing PWM motor control, encoder feedback, and closed-loop velocity control with <2% tracking error.
- Implemented an AprilTag-based vision alignment pipeline in Python/ROS2 for autonomous docking, integrating camera feedback with multi-axis actuation to achieve >95% docking success.

Arena Battle Robot, UPenn, Philadelphia, PA

Aug 2024 – Dec 2024

- Designed the robot chassis and drivetrain in SolidWorks, developing a compact, impact-resistant structure for high-load combat conditions.
- Built and assembled the system including component selection, perfboard soldering, and integration of motors, encoders, and power electronics.
- Implemented Vive-based localization and sensor fusion to estimate robot pose under dynamic motion and disturbances.
- Developed real-time control firmware in C++ on ESP32 for drive control, motor PWM, and failsafe handling, ensuring stable operation during competition.

Automated Adhesive Workstation, UofT, Toronto, Canada

Apr 2023 – Dec 2023

- Designed and assembled a pneumatic adhesive dispensing workstation in SolidWorks using machined aluminum components, linear slides, and press modules for automated production use.
- Integrated actuators, sensors, pneumatic system and PLC style control interfaces, enabling coordinated motion and closed-loop operation for dispensing tasks
- Performed system-level testing and iteration across 100+ cycles, improving accuracy to ± 0.2 mm and validating performance for production use.

INDUSTRY EXPERIENCES

Process Engineer, Bittelle Electronics Inc., Toronto, Canada

May 2022 – Sep 2023

- Designed test fixtures in SolidWorks for PCB and electromechanical assembly validation, improving repeatability and reducing manual setup errors.
- Built and optimized PCBA manufacturing and wiring harness processes, increasing production yield and improving assembly consistency.

Mechanical Engineer, Jiangnan Mould & Plastic Technology Co., Ltd, Shanghai, China

Apr 2021 – Aug 2021

- Designed and validated automotive components in CATIA using GD&T and tolerance analysis, ensuring manufacturability and robust assembly performance.
- Iterated through 10+ prototype builds, debugging structural issues using testing and CAE (Abaqus) to improve robustness.